

Troubleshooting VOIP & Network Connectivity Issues

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Purpose

This article walks front-line staff and IT support through diagnosing and resolving the most common VOIP phone and network connectivity issues encountered at station and corporate locations, before escalating to Tier 2 support.

Symptoms Covered

- No dial tone / phone shows “Registering” or “No Service”
- Calls dropping or cutting out
- One-way or garbled audio
- Phone loses connection intermittently

Before You Start

- Confirm the issue is isolated to one device, not a site-wide outage (check the IT status page / ask a nearby coworker).
- Have the phone's extension number and location on hand for the ticket.
- If this is a life-safety or urgent operational issue, escalate immediately rather than following this article.

Resolution Steps

1. Check physical connections. Verify the Ethernet cable is fully seated in both the phone and the wall/switch port. Swap the cable if it looks damaged.
2. Power-cycle the phone. Unplug it (or the PoE injector) for 15 seconds, then plug back in. Allow 60–90 seconds for it to fully register.
3. Verify network connectivity. From a laptop on the same port/switch, confirm you can reach the internet and internal resources. If the network itself is down, this is a network issue, not a phone issue — escalate to Network team.
4. Check for VLAN/port configuration issues. Confirm the switch port is assigned to the voice VLAN. A phone plugged into a data-only port will show “No Service.”
5. Test a loaner/known-good phone on the same port. If the loaner registers fine, the original handset is likely faulty — submit a hardware replacement ticket.
6. Review call quality settings. For one-way/garbled audio, check for bandwidth contention on the circuit (heavy Wi-Fi or large downloads on the same link can starve VOIP traffic, which relies on consistent low-latency bandwidth).
7. Document findings in the ticket, including which step resolved (or failed to resolve) the issue — this keeps the KB article accurate for the next technician.

Escalation Criteria

Escalate to Tier 2 / Network Engineering if:

- More than one phone at the same location is affected (possible switch or circuit issue).
- The issue persists after a confirmed known-good phone also fails on that port.
- The outage affects an operationally critical line (e.g., ramp, ops desk, or dispatch).

Related Articles

- [KB-2026-009: Diagnosing Wi-Fi Connectivity Drops at Station Locations](#)
- [KB-2026-011: Standard Ticket Escalation Matrix](#)

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